

# An Explainable Multimodal System for Stress Assessment and Emotion-Aware Supportive Dialogue

**Abstract**—Stress presents across physiological, speech, and linguistic channels, yet single-channel systems risk missing relevant cues, while naive fusion into one score obscures channel-specific differences. This paper presents an explainable multimodal framework that evaluates physiological, speech, and text channels independently before integrating their outputs into a shared state for supportive response generation. A wearable prototype (ESP32, MAX30102, CJMCU-6701) streams blood volume pulse and electrodermal activity to a server, where a calibrated logistic regression model trained on the WESAD dataset estimates stress probability. Speech is processed via a weighted ensemble trained on two acted corpora, with an appraisal rule mapping predicted emotion to a stress indicator. Text is analyzed using separate transformer classifiers for stress and cognitive distortion detection. The unified state drives emotion classification, next-turn intensity forecasting, rule-based strategy selection, and guarded response generation via a QLoRA-adapted Qwen3-4B model. The physiological model achieved macro-F1 of 0.866 under participant-independent evaluation; speech emotion recognition reached 0.527 under speaker-disjoint testing; text models achieved F1 of 0.808 (stress) and 0.755 (cognitive distortion). Each stage produces explanations via feature attributions, spectrogram maps, token attributions, or rule traces. Because no dataset jointly labels all constructs, and prototype sensor hardware differs from training-data sensors, results are reported per channel rather than fused. The intensity forecaster did not outperform a majority-class baseline. Future work includes validating sensors against reference recordings, evaluating speech-to-stress mapping with real stress labels, and improving forecasting with jointly labelled data.

**Keywords**—multimodal stress detection; speech emotion recognition; explainable artificial intelligence; probability calibration; emotional-support dialogue

## I. INTRODUCTION

Psychological stress is a complex state that leaves traces across several behavioural and physiological channels at once. These changes can be observed across multiple physiological and behavioral signals. In autonomic physiology, they may appear as variations in pulse-rate variability and skin conductance. In speech, they can be reflected through changes in arousal and prosodic features. They may emerge as characteristic patterns of appraisal and distorted reasoning. Over the course of a supportive conversation, these signals may also indicate changes in the type of response or support a person requires at a given moment. Each of these channels is noisy and each is partial, so a system that observes only one of them inherits the blind spots of that channel and cannot tell a missing signal from an absent state [1].

The usual response is to add modalities and then average them. This work takes the opposite position on the second half of that recipe. Combining physiological sensing with speech and text interpretation does supply complementary evidence, but merging it into one opaque score removes exactly what a supportive system needs: the physiological channel can indicate arousal while the language channel indicates that the person has already reappraised the situation, and those two facts should reach the response stage separately. Every modality here is therefore trained, evaluated and reported independently, and fused only into a small structured state whose fields remain individually inspectable.

That state drives a response pipeline that identifies the current emotional state, forecasts the intensity expected at

the following turn, selects a support strategy through deterministic rules, and generates a guarded reply. Explainability is attached at each stage rather than added as a display layer, so that predictions and strategy decisions can both be traced after the fact. The contributions of this work are as follows:

- An integrated multimodal architecture that combines continuous physiological sensing with speech and text-based analysis. The outputs from these modalities are mapped to a shared structured representation that supports emotion aware supportive response generation, with a defined input-output interface at each stage of the pipeline.
- A leakage controlled evaluation framework based on subject independent, speaker disjoint, and conversation disjoint validation protocols. The validation strategy is selected according to the primary leakage risk associated with each dataset and learning task.
- An explainable decision pipeline that combines feature level attribution, spectrogram and token level visualisation, and an auditable rule trace for support strategy selection. The framework also clearly distinguishes what each explanation method can and cannot establish.
- A wearable to application prototype that integrates physiological sensing, backend processing, speech and text analysis, multimodal state estimation, and strategy conditioned response generation within a deployed web application. The prototype also documents the components that remain experimental or require further validation.

## II. LITERATURE REVIEW

### A. Wearable Physiological Stress Sensing

Physiological stress detection in naturalistic settings was first explored in [2]. Since then, the Wearable Stress and Affect Detection dataset [3] has become a widely used benchmark, providing synchronised physiological recordings collected under a controlled experimental protocol. Because responses vary substantially between individuals, subject independent evaluation is essential: reviews report that a shared decision boundary across people is rarely appropriate [4] and that most published models are never shown to generalise to unseen participants [5]. Comparisons of user independent and personal models find neither uniformly superior [6], which is why this system pairs subject independent validation with a per participant baseline rather than choosing between them.

Measurement standards for both signals are established: acquisition and reporting requirements for electrodermal recording are set out in [7], continuous decomposition into tonic and phasic components in [8], and cardiac norms in [9], which establishes that the standard deviation of normal to normal intervals, the root mean square of successive differences and the proportion of successive intervals differing by more than fifty milliseconds are all estimable from roughly sixty seconds of data.

The deployment gap matters as much as the modelling: wrist estimates of pulse rate variability degrade under movement and speech [10], and optical pulse accuracy varies with device, activity and skin tone [11]. Because the

model trained here on research grade data is implemented on a low cost prototype, that difference is treated as the principal limitation rather than an implementation detail.

### B. Speech Emotion Recognition and Its Relation to Stress

Speech analysis has moved from classical acoustic descriptors, surveyed in [12] and standardised as a minimal parameter set in [13], towards self-supervised encoders such as wav2vec 2.0 [14] that learn representations directly from raw audio. The corpora used here, the Surrey Audio-Visual Expressed Emotion database [15] and the Ryerson Audio-Visual Database of Emotional Speech and Song [16], are both acted, which bounds what any result on them can mean.

Emotional state and stress are related but distinct constructs, so an emotion classifier cannot be read directly as a stress detector. Appraisal theory [17] relates emotional responses to perceived stress, and dimensional accounts of affect [18] supply the arousal axis along which that relation is expressed. This system therefore treats emotion recognition and stress estimation as separate stages, so that their assumptions can be evaluated independently and the mapping between them remains an explicit, inspectable rule.

### C. Text-Based Stress and Cognitive-Distortion Analysis

Transformer-based language models have improved text-based stress detection, with domain-adapted models such as MentalBERT [19] evaluated on datasets including Dreddit [20]. Cognitive distortion detection has been studied on annotated therapeutic and clinician patient exchanges [21], [22], and work on crowdsourced and online therapy text finds the binary decision far more tractable than fine grained type assignment [23], a finding adopted here by deploying the binary task and reserving the ten-label taxonomy for error analysis. Because these tasks are trained on different corpora, their outputs are not interchangeable, so the framework keeps the prediction stages separate and combines them only at fusion, where the combination is a pair of independent classifiers rather than a jointly supervised four class model.

### D. Emotional Support Dialogue and Efficient Adaptation

Research on emotional support dialogue has established the importance of response strategies, notably through the Emotional Support Conversation dataset [24], with related benchmarks addressing empathetic response [25] and emotion recognition in multi-party conversation [26]. Parameter efficient fine tuning, low rank adaptation [27] and its quantized variant [28], allows large language models to be adapted under limited computational budgets, and is used here to adapt Qwen3 [29] on consumer hardware; a graph based workflow framework [30] provides deterministic orchestration.

Safety is a first order concern rather than a deployment afterthought. An audit of widely used conversational agents found inconsistent recognition of crisis disclosures and inconsistent referral behaviour [31], which motivates routing high-risk inputs away from the generative model entirely rather than trusting it to behave well under distress.

### E. Explainability, Calibration and Evaluation

SHapley Additive Explanations (SHAP) assign additive feature contributions [32], Local Interpretable Model-Agnostic Explanations (LIME) fit a surrogate around one prediction [33], Integrated Gradients (IG) attribute an output along a baseline path [34], and Gradient-weighted Class Activation Mapping (Grad-CAM) localises evidence in feature maps [35]. All four describe model behaviour; agreement with human reasoning is a separate property that must be measured rather than assumed [36], and the case for preferring an interpretable model where one is competitive is argued in [37].

Calibration is likewise separate from ranking: high capacity classifiers are often overconfident, and post-hoc scaling improves reliability without changing the decision

order [38]. Evaluation under class imbalance also requires care, since accuracy and the F1 score can both be inflated when one class dominates; the Matthews correlation coefficient [39] and the precision recall curve [40] are more informative there, and standard treatments of imbalanced learning [41] frame the remedies. Comparing several models over multiple folds calls for non-parametric procedures [42], and topic modelling summarises recurring contexts without supervised labels [43].

Existing studies address physiological stress detection, speech emotion recognition, text-based stress analysis, cognitive distortion detection, emotional support dialogue and explainability as separate problems. The contribution here is therefore not a new model for an individual task but a unified architecture connecting these components into one pipeline in which each stage has a defined interface, is evaluated under the protocol appropriate to its own leakage risk, and retains its own probability, threshold and explanation. This lets physiological, vocal, textual and conversational evidence contribute jointly to an interpretable estimate of emotional state, and lets a weak component be identified as weak instead of hidden inside an aggregate score.

## III. SYSTEM ARCHITECTURE

Fig. 1 illustrates the architecture of the proposed multimodal stress and emotion aware support system. The system consists of four main stages: continuous physiological sensing, modality-specific interpretation, multimodal state fusion, and response generation.

A wearable prototype continuously acquires physiological signals and streams them to the backend. The physiological processing module maintains a calibrated stress probability, providing a continuous estimate of the user's stress state independently of the communication modality. Each conversational turn is then processed according to its input modality. For typed input, transformer based models independently detect stress and cognitive distortions, after which their outputs are combined by a four state late fusion scheme. For spoken input, an ensemble model identifies the user's emotional category, which an arousal gated appraisal rule subsequently converts into a stress indicator. Speech and text thus provide alternative interpretation pathways, while physiological information remains continuously available to both.

The outputs of the active conversational pathway are integrated with the current physiological estimate into a common structured representation holding a stress probability, an emotional category and a cognitive distortion state. This representation is the interface between the interpretation and response modules, so either side can be modified independently. The response module uses the shared state to classify the current emotional state, forecast the intensity expected at the subsequent turn, and select a support strategy through deterministic rules. A state machine determines the response pathway, routing high risk or crisis related inputs through a safeguarded pathway rather than to the generative model. For standard interactions a Qwen3-4B model adapted with quantized low rank adaptation generates the reply, which then passes through an output guard, and the complete pipeline is locally deployed research prototype web application.

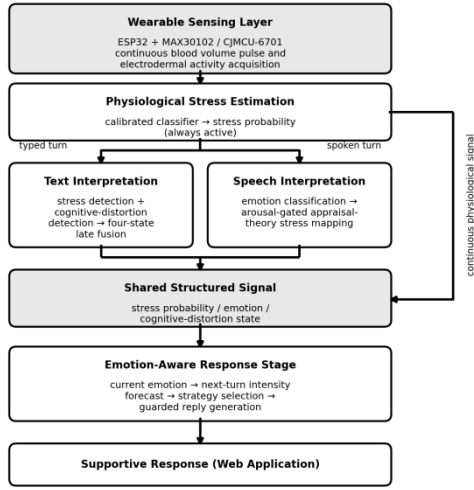


Fig. 1. Architecture of the proposed multimodal system. Continuous physiological sensing is integrated with alternative text and speech analysis pathways, with all modality specific outputs fused into a shared structured state for emotion aware response generation.

#### IV. SYSTEM IMPLEMENTATION

##### A. Physiological Sensing and Stress Estimation

The sensing stage detects stress from wrist blood volume pulse and electrodermal activity and targets a low cost prototype combining an ESP32 microcontroller with MAX30102 and CJMCU-6701 sensors. Models were trained and evaluated on the wrist data of the Wearable Stress and Affect Detection dataset [3] under a subject independent protocol, so that every reported score is obtained on participants the model has never seen. Fig. 2 shows the assembled prototype used for signal acquisition.

For each sixty second window, eighteen features describing pulse rate variability and skin conductance were extracted following the conventions of [7][9], and a five minute personal baseline normalises them so each value expresses deviation from that individual rather than from a population mean. Six approaches were compared, spanning classical models, a multilayer perceptron and a convolutional recurrent model (CRM). Logistic regression performed best, reaching a macro averaged F1 score of  $0.866 \pm 0.173$  and an area under the receiver operating characteristic curve of  $0.987 \pm 0.026$ , and non-parametric testing [42] confirmed its advantage over the weaker alternatives. The linear model is therefore among the best rather than uniquely best, consistent with preferring an interpretable model when it is competitive [37].

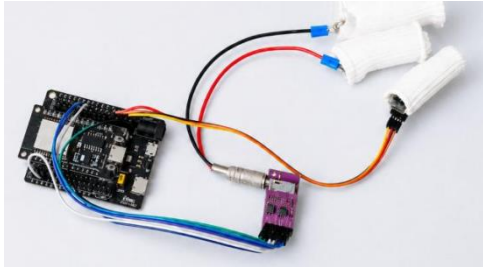


Fig. 2. Wearable sensing prototype. An ESP32 microcontroller acquires blood volume pulse data from a MAX30102 module and electrodermal activity from a CJMCU-6701 module, with electrodes and optical sensor mounted on fabric finger sleeves, and streams the raw signals to the backend for feature extraction and stress estimation.

The output is calibrated into a continuous stress probability consumed by later stages, and Shapley Additive Explanations (SHAP) [32] identify which features influenced individual predictions. The wearable only collects and streams signals; inference runs on the backend. Because the model was trained on research grade wrist

recordings rather than the prototype sensors, these results should not yet be read as validated prototype performance, for the reasons set out in [10][11].

##### B. Speech-Based Emotion Interpretation

For spoken turns the system first classifies the speaker's emotion and then converts it into a stress indicator using an arousal gated rule derived from appraisal theory [17] and the dimensional account of affect in [18], keeping emotion recognition separate from stress interpretation and making the mapping between them an explicit object that can be replaced or audited.

Seven approaches were compared on the combined Surrey Audio-Visual Expressed Emotion [15] and Ryerson Audio-Visual Database of Emotional Speech and Song [16] corpora, comprising 1,727 samples from 28 speakers across seven emotions, spanning traditional acoustic methods [12], [13], deep learning models, a finetuned wav2vec 2.0 encoder [14] and a weighted ensemble. A speaker independent 70/15/15 split prevented any speaker from appearing in more than one partition.

The weighted ensemble of a support vector machine with a radial basis function kernel, a convolutional recurrent network and the finetuned encoder achieved the best result, with a macro averaged F1 score of 0.5274, an accuracy of 0.5399, an area under the receiver operating characteristic curve of 0.8616 and an expected calibration error of 0.0640. The finetuned encoder alone reached a macro averaged F1 score of 0.5013. Grad-CAM [35] highlights the time-frequency regions influencing spectrogram-based predictions. The resulting emotion prediction is converted into a stress indicator for the multimodal state.

##### C. Text Based Stress and Cognitive Distortion Analysis

For typed input the system analyses free text, such as diary entries, for stress and for cognitive distortions. Three transformer models BERT [44], MentalBERT [19] and DeBERTa-v3 [45] were finetuned separately for stress detection on Dreddit [20] and for cognitive distortion detection on a therapist question and answer corpus [22]. The source cognitive distortion dataset contains ten named distortion types, but the deployed cognitive distortion task is binary: No Distortion versus Distortion. MentalBERT, the domain adapted encoder, achieved the best individual performance, with a macro averaged F1 score of 0.8077 for stress and 0.7554 for cognitive distortion. Weighted ensembles were also evaluated for both tasks. The stress ensemble combines BERT, MentalBERT, and DeBERTa-v3 with weights of 0.50, 0.10, and 0.40, respectively, while the cognitive distortion ensemble combines BERT, MentalBERT and DeBERTa-v3 with weights 0.10, 0.40 and 0.50, respectively. Integrated Gradients (IG) [34], Shapley Additive Explanations (SHAP) [32] and Local Interpretable Model-Agnostic Explanations (LIME) [33] explain model decisions, while unsupervised topic modelling [43] identifies recurring stress and distortion themes.

The stress ensemble uses a raw decision threshold of 0.53 and a calibrated threshold of 0.4195, while the cognitive distortion ensemble uses a calibrated threshold of 0.4861. Affine temperature scaling improved expected calibration error from 0.0694 to 0.0625 for stress detection and from 0.1296 to 0.0775 for cognitive distortion detection. Their binary outputs are combined by decision level late fusion into four states: Neither, Distortion only, Stress only, or Both. Because no corpus carries both labels on the same sample, this is a combination of two independently trained classifiers rather than a jointly supervised four class model, and no four class accuracy is reported. The ten named distortion types are retained only to diagnose binary errors and do not constitute ten way performance, consistent with the difficulty of fine grained assignment reported in [23]. The text analysis module is integrated into the wider system through a web service interface.

#### D. Emotion Aware Response Generation

The response module follows a forecast then respond design. It identifies the current emotion, forecasts the intensity expected at the following turn, selects a support strategy through predefined rules, and generates a reply conditioned on that strategy. A graph based workflow [30] coordinates these steps and enforces safety controls that route crisis related inputs away from the language model and supply fallback responses when required, following the failure modes documented in [31].

RoBERTa [46] serves as the current emotion classifier, reaching 82.23% accuracy and a macro average F1 score of 0.8171 on 1,030 held out utterances. The forecasting stage does not predict the next emotion category. It predicts the intensity of the emotion expected at the following turn as a binary low or high label, which is the quantity the strategy selection rules actually consume. The deployed forecaster is a hybrid that combines a sparse lexical baseline with a transformer fusion model using equal weights, so the final probability is the mean of the two, with a decision threshold of 0.595 selected on the validation split. Both components are trained and evaluated on the Emotional Support Conversation dataset [24].

On 200 held out test turns the hybrid model classified 158 correctly, giving 79.0 percent accuracy. This is below the 84.0 percent obtained by always predicting the majority low class, and the area under the receiver operating characteristic curve of 0.576 indicates only a weak ability to rank high intensity turns above low intensity ones against the 0.50 expected from random ranking. The 42 errors comprise 29 high intensity turns classified as low and 13 low intensity turns classified as high, so only 3 of the 32 high intensity cases were recovered. The transformer fusion model alone is more useful for the minority class, reaching 28.13% high intensity recall and a high class F1 score of 0.231. Per turn outputs recording the actual intensity, the predicted intensity and the predicted probability of high intensity are retained for error analysis, and the full metric set appears in Table II.

For response generation, Qwen3 [29] was adapted using quantized low rank adaptation [28] and finetuned on the Emotional Support Conversation dataset [24] with only 33 million trainable parameters. The adapted model produced shorter supportive replies, reducing average length from 56.6 to 12.0 words. It reached an evaluation loss of 2.091 and a perplexity of 8.09 while requiring 3.05 gigabytes of peak graphics processing unit memory, which supports deployment on consumer hardware.

### V. RESULTS AND DISCUSSION

Table I summarises the final performance of each component. Because the text analysis and response modules contain several tasks, the table reports the best or reference result for each task rather than one overall score, and each result should be read together with its dataset and evaluation protocol. For the two text tasks it lists both the strongest individual model and the deployed ensemble, so that the two can be compared directly. The system does not produce a single end to end accuracy, and combining the individual scores into one figure would be misleading rather than informative.

TABLE I. FINAL RESULTS BY PROCESSING STAGE

| Processing stage      | Task                   | Model               | Result                                                                                                    |
|-----------------------|------------------------|---------------------|-----------------------------------------------------------------------------------------------------------|
| Physiological sensing | Stress detection       | Logistic regression | Macro-averaged F1 0.866; area under receiver operating characteristic curve 0.987 (leave one subject out) |
| Speech pathway        | Emotion classification | Weighted ensemble   | Accuracy 0.5399; macro-averaged F1 0.5274; area under curve 0.8616                                        |
| Text pathway          | Stress detection       | MentalBERT          | Accuracy 0.8098; macro-averaged F1 0.8077                                                                 |

|                |                      |                                                  |                                                                    |
|----------------|----------------------|--------------------------------------------------|--------------------------------------------------------------------|
| Text pathway   | Distortion detection | MentalBERT                                       | Accuracy 0.7747; macro-averaged F1 0.7554                          |
| Text pathway   | Stress detection     | BERT + MentalBERT + DeBERTa-v3 weighted ensemble | Accuracy 0.8070; macro-F1 0.8062; AUROC 0.8971                     |
| Text pathway   | Distortion detection | BERT + MentalBERT + DeBERTa-v3 weighted ensemble | Accuracy 0.7648; macro-averaged F1 0.7425; area under curve 0.8490 |
| Response stage | Current emotion      | RoBERTa                                          | Accuracy 0.8223; macro-averaged F1 0.8171                          |
| Response stage | Next-turn intensity  | Hybrid (sparse + transformer)                    | Accuracy 0.790; macro-averaged F1 0.503; area under curve 0.576    |
| Response stage | Reply generation     | Qwen3-4B with quantized low-rank adaptation      | Evaluation loss 2.091; perplexity 8.09                             |

#### A. Evaluation Protocols

Performance is reported separately for each stage and task because the system spans different modalities, datasets, labels and measures. The physiological model is evaluated across subjects, the speech models across speakers, the text models on held out documents, and the response module across conversations, each protocol targeting the dominant leakage risk for its modality: subject overlap, speaker overlap and dialogue overlap respectively. These protocols give a more reliable estimate of within dataset generalisation, but establish nothing about generalisation across datasets or to real world conditions, and the systematic review in [5] indicates that this is the distinction most often collapsed in published work.

#### B. Interpretation of Results

Each result should be read against the task actually evaluated at that stage. Speech performance reflects emotion recognition, not the accuracy of the stress indicator produced by the subsequent appraisal mapping, which has no direct ground truth. The next turn intensity forecaster did not exceed the majority low baseline and recovered only 3 of 32 high intensity turns; under this imbalance accuracy is the least informative available measure, and the Matthews correlation coefficient of 0.022 and the area under the precision recall curve of 0.197 in Table II are the numbers that characterise the component honestly [39], [40]. It has therefore not demonstrated a practical advantage over predicting the majority class, and it is reported rather than removed because a forecasting stage that fails on the minority class is a result the field needs and rarely sees. The text ensembles improved some ranking metrics without consistently beating the strongest individual models on accuracy, which is why both are reported side by side.

TABLE II. NEXT-TURN INTENSITY FORECASTING ON 200 HELD-OUT CONVERSATION TURNS

| Metric                                             | Hybrid forecaster |
|----------------------------------------------------|-------------------|
| Accuracy                                           | 0.790             |
| Balanced accuracy                                  | 0.508             |
| Macro-averaged F1                                  | 0.503             |
| High-intensity precision                           | 0.188             |
| High-intensity recall                              | 0.094             |
| High-intensity F1                                  | 0.125             |
| Area under receiver operating characteristic curve | 0.576             |
| Area under precision-recall curve                  | 0.197             |
| Matthews correlation coefficient                   | 0.022             |
| Majority-class baseline accuracy                   | 0.840             |

The complete pipeline has been functionally integrated and locally deployed research prototype web application. Physiological stress is supplied continuously from the backend while each turn is processed through the appropriate speech or text pathway, and the result is converted into the common structured representation that

guides response generation. Performance nevertheless remains reported at the stage level, since the components address different tasks on different corpora and no suitable jointly labelled dataset exists.

### C. Explainability and Ensembling

Interpretability methods were selected by model type. Shapley Additive Explanations (SHAP) [32] explain physiological predictions through the contribution of engineered features, and Gradient-weighted Class Activation Mapping (Grad-CAM) [35] identifies the time frequency regions influencing speech predictions. For text, Integrated Gradients (IG) [34], Shapley Additive Explanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME) [33] give qualitative accounts of model decisions without any claim of agreement with human annotated rationales, a property that would have to be measured separately [36]. These three methods are applied to every deployed ensemble member, and the resulting attributions are normalised and aggregated using the same weights used for prediction, so the explanation presented for a text decision corresponds to the ensemble that produced it. In the response stage an auditable rule trace records how the strategy was selected and what the output guard did. It explains the decision, not the individual tokens the language model produced.

Ensembling was used differently across modalities. The speech ensemble combines the support vector machine, the convolutional recurrent network and the finetuned encoder with validation derived weights of 0.0, 0.2 and 0.8, showing that the final ensemble relies almost entirely on the neural components and that the classical model contributes nothing at the operating point. For text, separate ensembles serve the two tasks. For stress detection, the ensemble combines BERT, MentalBERT, and DeBERTa-v3 with weights of 0.50, 0.10, and 0.40, respectively, achieving an accuracy of 0.8070, a macro-averaged F1 score of 0.8062, and an AUROC of 0.8971. For cognitive distortion detection the ensemble combines BERT, MentalBERT and DeBERTa-v3 with weights of 0.10, 0.40 and 0.50 and a decision threshold of 0.4861, and it achieved an accuracy of 0.7648, a macro averaged F1 score of 0.7425 and an area under the receiver operating characteristic curve of 0.8490. Ensemble results are reported alongside the strongest individual models rather than on the assumption that ensembling always helps.

## VI. LIMITATIONS AND FUTURE WORK

### A. Domain Mismatch and Missing Ground Truth

The principal limitation is the mismatch between the training and deployment domains. The physiological model is trained on research-grade wrist recordings, whereas the prototype uses MAX30102 optical pulse data and CJMCU-6701 readings on an analogue-to-digital converter scale. Equivalent modalities do not necessarily provide equivalent transfer functions, units, placement, contact pressure, or noise characteristics; published evidence shows that agreement varies with posture, movement, and speech [10], while optical accuracy can vary with device, activity, and skin tone [11]. The speech models are evaluated on acted corpora [15], [16], the text models on social media and therapist data rather than private diaries, and the forecaster on a crowdsourced corpus [24]. Therefore, the reported results are dataset-specific and should not be interpreted as clinical or unrestricted real-world performance.

Several outputs also lack direct ground truth. Speech is evaluated for emotion rather than labelled stress, while the four-state text fusion combines independently trained classifiers because no corpus provides both labels for the same sample. Consequently, no end-to-end accuracy is reported. Explainability coverage is also limited: the text explanations have not been validated against human rationales, and the response-stage rule trace explains

strategy selection rather than the internal behaviour of the language model. Generated responses are evaluated only using automatic metrics, without human, expert, or clinical assessment.

### B. Privacy, Safety and Human-Centred Use

Both the sensing and language components process sensitive wellbeing information. The language component stores diary text and model outputs locally, so any production deployment requires retention limits, encryption at rest and in transit, role based access, deletion controls, informed consent and audit logging. The framework must also communicate its scope plainly. Physiological arousal is not psychological stress, an emotion label is not a diagnosis, and a distortion signal is not a clinical finding. Messages should preserve component level probabilities and explanations, allow an interpretation to be dismissed, and never serve as the sole basis for a clinical, educational or employment decision. The crisis pathway is deliberately deterministic and placed ahead of the generative model, because conversational agents have been shown to respond inconsistently to disclosures of distress [31].

### C. Future Work

The next stage is to convert repository level evidence into deployment evidence rather than to expand the model list. A controlled prototype study should collect synchronised prototype and reference recordings, log placement, acquisition jitter, contact quality and raw units, and compare participant level performance and calibration before and after transfer. Speech stress should be evaluated against direct stress labels so the appraisal mapping is measured rather than assumed, and text fusion tested on a jointly labelled corpus. Next turn intensity forecasting needs explicit class imbalance handling [41], richer conversational context and additional corpora such as [25][26]. Human and expert evaluation of response quality and safety should accompany system level measurement of latency, reliability and consistency.

## VII. CONCLUSION

This research presented an integrated and explainable system that combines physiological, speech and text evidence to assess stress and emotion and to generate supportive replies, and that keeps those channels separate on purpose. The physiological model reached a leave one, subject out, macro averaged F1 score of 0.866, speech emotion recognition reached 0.5274 under a speaker disjoint split, and the text models reached 0.8077 for stress and 0.7554 for cognitive distortion. These outputs are combined into a shared structured representation supporting current emotion classification, next turn intensity forecasting, strategy selection and guarded response generation from a language model adapted with quantized low rank adaptation. Explainability is provided through feature and token attribution, spectrogram localisation and auditable decision rules, and the complete pipeline was locally deployed research prototype web application. The evaluation is component level by design, because low cost sensor transfer, the emotion to stress mapping and jointly labelled multimodal ground truth all remain unvalidated, and because the next turn intensity forecaster does not yet beat its majority class baseline. Establishing those three things, rather than adding further models, is the work that would make the system deployable.

## REFERENCES

- [1] World Health Organization, "Mental disorders," World Health Organization, Geneva, Switzerland, Sep. 30, 2025. Accessed: Aug. 19, 2026. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/mental-disorders>
- [2] J. A. Healey and R. W. Picard, "Detecting stress during real-world driving tasks using physiological sensors," *IEEE Trans. Intell. Transp. Syst.*, vol. 6, no. 2, pp. 156–166, 2005, doi: 10.1109/TITS.2005.848368.

- [3] P. Schmidt, A. Reiss, R. Duerichen, C. Marberger, and K. Van Laerhoven, "Introducing WESAD, a multimodal dataset for wearable stress and affect detection," in *Proc. 20th ACM Int. Conf. Multimodal Interact.*, Boulder, CO, USA, 2018, pp. 400–408, doi: 10.1145/3242969.3242985.
- [4] Y. S. Can, B. Amrich, and C. Ersoy, "Stress detection in daily life scenarios using smart phones and wearable sensors: A survey," *J. Biomed. Inform.*, vol. 92, p. 103139, 2019, doi: 10.1016/j.jbi.2019.103139.
- [5] G. Vos, K. Trinh, Z. Sarnyai, and M. Rahimi Azghadi, "Generalizable machine learning for stress monitoring from wearable devices: A systematic literature review," *Int. J. Med. Inform.*, vol. 173, p. 105026, 2023, doi: 10.1016/j.ijmedinf.2023.105026.
- [6] P. Siirtola and J. Rönning, "Comparison of regression and classification models for user-independent and personal stress detection," *Sensors*, vol. 20, no. 16, p. 4402, 2020, doi: 10.3390/s20164402.
- [7] W. Boucsein et al., "Publication recommendations for electrodermal measurements," *Psychophysiology*, vol. 49, no. 8, pp. 1017–1034, 2012, doi: 10.1111/j.1469-8986.2012.01384.x.
- [8] M. Benedek and C. Kaernbach, "A continuous measure of phasic electrodermal activity," *J. Neurosci. Methods*, vol. 190, no. 1, pp. 80–91, 2010, doi: 10.1016/j.jneumeth.2010.04.028.
- [9] F. Shaffer and J. P. Ginsberg, "An overview of heart rate variability metrics and norms," *Front. Public Health*, vol. 5, p. 258, 2017, doi: 10.3389/fpubh.2017.00258.
- [10] L. Menghini et al., "Stressing the accuracy: Wrist-worn wearable sensor validation over different conditions," *Psychophysiology*, vol. 56, no. 11, p. e13441, 2019, doi: 10.1111/psyp.13441.
- [11] B. Bent, B. A. Goldstein, W. A. Kibbe, and J. P. Dunn, "Investigating sources of inaccuracy in wearable optical heart rate sensors," *npj Digit. Med.*, vol. 3, p. 18, 2020, doi: 10.1038/s41746-020-0226-6.
- [12] M. El Ayadi, M. S. Kamel, and F. Karray, "Survey on speech emotion recognition: Features, classification schemes, and databases," *Pattern Recognit.*, vol. 44, no. 3, pp. 572–587, 2011, doi: 10.1016/j.patcog.2010.09.020.
- [13] F. Eyben et al., "The Geneva minimalistic acoustic parameter set for voice research and affective computing," *IEEE Trans. Affective Comput.*, vol. 7, no. 2, pp. 190–202, 2016, doi: 10.1109/TAFFC.2015.2457417.
- [14] A. Baevski, Y. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 12449–12460.
- [15] S. Haq and P. J. B. Jackson, "Multimodal emotion recognition," in *Machine Audition: Principles, Algorithms and Systems*, W. Wang, Ed. Hershey, PA, USA: IGI Global, 2010, pp. 398–423, doi: 10.4018/978-1-61520-919-4.ch017.
- [16] S. R. Livingstone and F. A. Russo, "The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS): A dynamic, multimodal set of facial and vocal expressions in North American English," *PLoS ONE*, vol. 13, no. 5, p. e0196391, 2018, doi: 10.1371/journal.pone.0196391.
- [17] R. S. Lazarus and S. Folkman, *Stress, Appraisal, and Coping*. New York, NY, USA: Springer, 1984.
- [18] J. A. Russell, "A circumplex model of affect," *J. Pers. Soc. Psychol.*, vol. 39, no. 6, pp. 1161–1178, 1980, doi: 10.1037/h0077714.
- [19] S. Ji et al., "MentalBERT: Publicly available pretrained language models for mental healthcare," in *Proc. 13th Lang. Resour. Eval. Conf.*, Marseille, France, 2022, pp. 7184–7190.
- [20] E. Turcan and K. McKeown, "Dreaddit: A Reddit dataset for stress analysis in social media," in *Proc. 10th Int. Workshop Health Text Mining Inf. Anal.*, Hong Kong, China, 2019, pp. 97–107, doi: 10.18653/v1/D19-6213.
- [21] J. S. Tauscher et al., "Automated detection of cognitive distortions in text exchanges between clinicians and people with serious mental illness," *Psychiatr. Serv.*, vol. 74, no. 4, pp. 407–410, 2023, doi: 10.1176/appi.ps.202100692.
- [22] S. Shreevastava and P. Foltz, "Detecting cognitive distortions from patient-therapist interactions," in *Proc. 7th Workshop Comput. Linguistics Clin. Psychol.*, 2021, pp. 151–158, doi: 10.18653/v1/2021.clpsych-1.17.
- [23] B. Shickel et al., "Automatic detection and classification of cognitive distortions in mental health text," in *Proc. IEEE 20th Int. Conf. Bioinformatics Bioeng.*, 2020, pp. 275–280, doi: 10.1109/BIBE50027.2020.00052.
- [24] S. Liu et al., "Towards emotional support dialog systems," in *Proc. 59th Annu. Meeting Assoc. Comput. Linguistics 11th Int. Joint Conf. Natural Lang. Process.*, 2021, pp. 3469–3483, doi: 10.18653/v1/2021.acl-long.269.
- [25] H. Rashkin, E. M. Smith, M. Li, and Y.-L. Boureau, "Towards empathetic open-domain conversation models: A new benchmark and dataset," in *Proc. 57th Annu. Meeting Assoc. Comput. Linguistics*, Florence, Italy, 2019, pp. 5370–5381, doi: 10.18653/v1/P19-1534.
- [26] S. Poria et al., "MELD: A multimodal multi-party dataset for emotion recognition in conversations," in *Proc. 57th Annu. Meeting Assoc. Comput. Linguistics*, Florence, Italy, 2019, pp. 527–536, doi: 10.18653/v1/P19-1050.
- [27] E. J. Hu et al., "LoRA: Low-rank adaptation of large language models," in *Proc. 10th Int. Conf. Learn. Representations*, 2022.
- [28] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient finetuning of quantized large language models," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 36, New Orleans, LA, USA, 2023, pp. 10088–10115.
- [29] Qwen Team, "Qwen3 technical report," Alibaba Group, Hangzhou, China, Tech. Rep., 2025.
- [30] LangChain, Inc., "LangGraph overview," LangChain Documentation. Accessed: Aug. 19, 2026. [Online]. Available: <https://docs.langchain.com/oss/python/langgraph/overview>
- [31] A. S. Miner, A. Milstein, S. Schueller, R. Hegde, C. Mangurian, and E. Linos, "Smartphone-based conversational agents and responses to questions about mental health, interpersonal violence, and physical health," *JAMA Intern. Med.*, vol. 176, no. 5, pp. 619–625, 2016, doi: 10.1001/jamainternmed.2016.0400.
- [32] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Long Beach, CA, USA, 2017, pp. 4765–4774.
- [33] M. T. Ribeiro, S. Singh, and C. Guestrin, "Why should I trust you? Explaining the predictions of any classifier," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, San Francisco, CA, USA, 2016, pp. 1135–1144, doi: 10.1145/2939672.2939778.
- [34] M. Sundararajan, A. Taly, and Q. Yan, "Axiomatic attribution for deep networks," in *Proc. 34th Int. Conf. Mach. Learn.*, PMLR, vol. 70, 2017, pp. 3319–3328.
- [35] R. R. Selvaraju et al., "Grad-CAM: Visual explanations from deep networks via gradient-based localization," *Int. J. Comput. Vis.*, vol. 128, no. 2, pp. 336–359, 2020, doi: 10.1007/s11263-019-01228-7.
- [36] A. Jacovi and Y. Goldberg, "Towards faithfully interpretable NLP systems: How should we define and evaluate faithfulness?" in *Proc. 58th Annu. Meeting Assoc. Comput. Linguistics*, 2020, pp. 4198–4205, doi: 10.18653/v1/2020.acl-main.386.
- [37] C. Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead," *Nat. Mach. Intell.*, vol. 1, no. 5, pp. 206–215, 2019, doi: 10.1038/s42256-019-0048-x.
- [38] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. 34th Int. Conf. Mach. Learn.*, PMLR, vol. 70, 2017, pp. 1321–1330.
- [39] D. Chicco and G. Jurman, "The advantages of the Matthews correlation coefficient over F1 score and accuracy in binary classification evaluation," *BMC Genomics*, vol. 21, no. 1, p. 6, 2020, doi: 10.1186/s12864-019-6413-7.
- [40] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," *PLoS ONE*, vol. 10, no. 3, p. e0118432, 2015, doi: 10.1371/journal.pone.0118432.
- [41] H. He and E. A. Garcia, "Learning from imbalanced data," *IEEE Trans. Knowl. Data Eng.*, vol. 21, no. 9, pp. 1263–1284, 2009, doi: 10.1109/TKDE.2008.239.
- [42] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," *J. Mach. Learn. Res.*, vol. 7, pp. 1–30, 2006.
- [43] R. Egger and J. Yu, "A topic modeling comparison between LDA, NMF, Top2Vec, and BERTopic to demystify Twitter posts," *Front. Sociol.*, vol. 7, p. 886498, 2022, doi: 10.3389/fsoc.2022.886498.
- [44] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics: Human Lang. Technol.*, Minneapolis, MN, USA, 2019, pp. 4171–4186, doi: 10.18653/v1/N19-1423.
- [45] P. He, J. Gao, and W. Chen, "DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing," in *Proc. 11th Int. Conf. Learn. Representations*, Kigali, Rwanda, 2023.
- [46] Y. Liu et al., "RoBERTa: A robustly optimized BERT pretraining approach," Facebook AI Research and Paul G. Allen School Comput. Sci. Eng., Univ. Washington, Seattle, WA, USA, Tech. Rep., 2019.